

NEXUSPLATFORM XR-7000 ? PRODUCT MANUAL

Product Overview

The NexusPlatform XR-7000 is an enterprise-grade data integration and analytics platform designed for large-scale organizations. Model identifier: NP-XR-7000-ENT.

Key Capabilities:

- Real-time data ingestion from 200+ source connectors.
- Built-in ETL pipeline builder with drag-and-drop interface.
- Advanced analytics engine with SQL and Python support.
- Role-based access control with SSO integration.
- Horizontal scaling across distributed clusters.

Product Editions:

- XR-7000 Standard: Up to 100 concurrent users.
- XR-7000 Professional: Up to 500 concurrent users with premium connectors.
- XR-7000 Enterprise: Unlimited users, dedicated support, custom SLA.

This manual covers installation, configuration, operation, and troubleshooting for all editions of the XR-7000 platform.

System Requirements

Minimum Hardware Requirements:

- CPU: 8 cores (Intel Xeon or AMD EPYC recommended).
- RAM: 32 GB minimum, 64 GB recommended for production.
- Storage: 500 GB SSD with minimum 3000 IOPS.
- Network: 1 Gbps Ethernet interface.

Supported Operating Systems:

- Ubuntu 20.04 LTS or later.
- Red Hat Enterprise Linux 8.x or later.
- CentOS Stream 8 or later.
- Windows Server 2019 or later (limited support).

Software Prerequisites:

- Java Runtime Environment (JRE) 17 or later.
- PostgreSQL 14 or later (for metadata store).
- Docker 24.0 or later (for containerized deployment).
- Python 3.10 or later (for custom analytics scripts).

Browser Compatibility (Web Dashboard):

- Google Chrome 100+
- Mozilla Firefox 100+
- Microsoft Edge 100+
- Safari 15+

Installation Guide

Step 1: Download the installer package from the NexusPlatform portal.

Step 2: Verify the checksum: `sha256sum nexusplatform-xr7000-latest.tar.gz`

Step 3: Extract the archive: `tar -xzf nexusplatform-xr7000-latest.tar.gz`

Step 4: Run the installer: `./install.sh --edition enterprise`

Step 5: Follow the interactive setup wizard to configure database connection, admin credentials, and license key.

Initial Configuration:

After installation, access the web dashboard at <https://localhost:8443> with the admin credentials created during setup.

Default Ports:

- Web Dashboard: 8443 (HTTPS)
- API Gateway: 9090
- Data Ingestion: 9091
- Metrics: 9092

License Activation:

Enter your license key in Settings > License Management. The platform operates in trial mode (30 days, 10 users) without a license key.

API Configuration and Authentication

API Authentication Tokens:

The XR-7000 API uses bearer token authentication. To generate an API token:

1. Navigate to Settings > API Management > Tokens.
2. Click "Generate New Token."
3. Select the permission scope (read-only, read-write, or admin).
4. Set an expiration period (30, 90, 180, or 365 days).
5. Copy the generated token immediately ? it will not be shown again.

API Rate Limits:

- Standard edition: 100 requests per minute per token.
- Professional edition: 500 requests per minute per token.
- Enterprise edition: 2000 requests per minute per token.

API Base URL:

`https://<your-instance>/api/v2/`

Example API Call:

```
curl -H "Authorization: Bearer <token>" https://instance.example.com/api/v2/datasets
```

Webhook Configuration:

Configure webhooks in Settings > Integrations > Webhooks to receive real-time notifications for data pipeline events, alerts, and system status changes.

Error Codes and Troubleshooting

Common Error Codes:

E100 - Connection Timeout: The platform could not reach the data source within the configured timeout period. Check network connectivity and firewall rules.

E101 - Authentication Failed: Invalid credentials for the data source connector. Verify username, password, or API key.

E102 - Schema Mismatch: The source data schema does not match the expected target schema. Review field mappings in the pipeline configuration.

E103 - Storage Full: The storage volume has reached capacity. Free disk space or expand the volume.

E104 - License Expired: The platform license has expired. Contact sales@nexusplatform.com to renew. The platform continues in read-only mode for 7 days after expiration.

E105 - Rate Limit Exceeded: API rate limit has been exceeded. Wait for the rate limit window to reset or upgrade your edition.

Reset Procedure:

To reset the platform to factory defaults:

1. Stop all services: `nexusctl stop --all`
2. Run the reset command: `nexusctl reset --factory --confirm`
3. This removes all data, configurations, and user accounts.
4. Re-run the installation wizard after reset.

WARNING: Factory reset is irreversible. Create a backup before proceeding.

Network Requirements and Connectivity

Network Architecture:

The XR-7000 requires the following network configuration for proper operation:

Required Ports (Inbound):

- TCP 8443: Web dashboard (HTTPS).
- TCP 9090: API gateway.
- TCP 9091: Data ingestion endpoint.

Required Ports (Outbound):

- TCP 443: License server, update server, cloud connectors.
- TCP 5432: PostgreSQL metadata store (if external).
- TCP/UDP as required by configured data source connectors.

Firewall Requirements:

- Allow outbound HTTPS to `license.nexusplatform.com` for license validation.
- Allow outbound HTTPS to `updates.nexusplatform.com` for software updates.
- Internal cluster nodes must have unrestricted communication on ports 9100-9110.

Load Balancer Configuration:

For high-availability deployments, configure a Layer 7 load balancer with:

- Health check endpoint: `GET /api/v2/health`
- Session affinity: enabled (cookie-based).
- SSL termination at load balancer recommended.

Bandwidth Recommendations:

- Minimum: 1 Gbps for standard workloads.
- Recommended: 10 Gbps for high-throughput data ingestion scenarios.

Service Level Agreement and Warranty

SLA Uptime Guarantee:

The XR-7000 Enterprise edition includes a 99.9% uptime SLA for the managed cloud deployment option. This translates to a maximum of 8.76 hours of unplanned downtime per year. Scheduled maintenance windows are excluded from SLA calculations.

SLA Credits:

- 99.0% - 99.9% uptime: 10% service credit.
- 95.0% - 99.0% uptime: 25% service credit.
- Below 95.0% uptime: 50% service credit.

Support Response Times:

- Critical (P1): 1 hour response, 4 hour resolution target.
- High (P2): 4 hour response, 1 business day resolution target.
- Medium (P3): 1 business day response.
- Low (P4): 2 business day response.

Warranty:

The XR-7000 hardware appliance (if purchased) includes a 3-year limited warranty covering manufacturing defects. The warranty does not cover damage from improper installation, unauthorized modifications, or environmental factors. Software updates are included for the duration of the active subscription.

Extended warranty and premium support plans are available. Contact your account manager for details.

Enterprise Subscription Plans and Pricing

Subscription Tiers:

- Standard: \$500/month ? up to 100 users, 1 TB storage, email support.
- Professional: \$1,500/month ? up to 500 users, 5 TB storage, priority support.
- Enterprise: Custom pricing ? unlimited users, unlimited storage, dedicated support, custom SLA.

All subscriptions are billed annually. Monthly billing is available at a 15% premium.

Refund Policy for Enterprise Subscriptions:

Enterprise subscription refunds are processed under the following conditions:

1. Cancellation within 30 days of initial purchase: full refund.
2. Cancellation within 31-90 days: pro-rated refund minus a 15% administrative fee.
3. Cancellation after 90 days: no refund; service continues until the end of the billing period.
4. Refund requests must be submitted in writing to billing@nexusplatform.com.
5. Custom development work and professional services fees are non-refundable.

Subscription Renewal:

Subscriptions auto-renew 30 days before expiration. Customers are notified 60 days in advance. To cancel auto-renewal, submit a request at least 45 days before the renewal date.